

Mistral Dynamic Agent Tools Project File Report

Generated: 13 May 2026, 17:07 | Total files: 113

■ Root / Project-Level Files

File	Description	APIs / Endpoints	LOC
LICENSE LICENSE	Project support file.	—	21
README.md README.md	Project-level README with setup, architecture overview, and usage instructions.	—	262
api_documentation.md api_documentation.md	Human-readable API reference summarising all backend routes.	—	120
hybrid_architecture_implementation.md hybrid_architecture_implementation.md	Deep-dive implementation notes on the hybrid Mistral + local workflow architecture.	—	925
mistral-agent-frontend-spec.md mistral-agent-frontend-spec.md	Markdown documentation file.	—	2625

■ Backend Service (FastAPI)

File	Description	APIs / Endpoints	LOC
_diag.py backend/_diag.py	Diagnostic script for inspecting backend state and DB contents.	—	33

File	Description	APIs / Endpoints	LOC
check_stuck.py backend/check_stuck.py	Utility to detect and report workflow tasks stuck in a pending state.	—	24
check_tasks.py backend/check_tasks.py	Utility to list and inspect current background tasks.	—	18
direct_recompile.py backend/direct_recompile.py	Script to force-recompile a workflow module without restarting the server.	—	28
re_register.py backend/re_register.py	Script to re-register workflow definitions with the Mistral cloud.	—	13
requirements.txt backend/requirements.txt	Python package dependencies for the service.	—	10
rewrite.py backend/rewrite.py	Utility to rewrite/migrate tool or workflow records in the database.	—	144
test_endpoint.py backend/test_endpoint.py	Ad-hoc HTTP test script for manually probing backend endpoints.	—	19
test_stream.py backend/test_stream.py	Ad-hoc test script for validating SSE streaming from the backend.	—	23
__init__.py backend/app/__init__.py	Python module providing supporting logic for the service.	—	1
config.py backend/app/config.py	Loads and exposes environment-level configuration (API keys, DB URL, etc.).	—	54
database.py backend/app/database.py	SQLAlchemy engine / session factory and Base declarative model.	—	36
dependencies.py backend/app/dependencies.py	FastAPI dependency-injection helpers (DB session, auth tokens, etc.).	—	29
exceptions.py backend/app/exceptions.py	Custom HTTP exception classes and global exception handlers.	—	90

File	Description	APIs / Endpoints	LOC
main.py backend/app/main.py	Application entry point; registers routers and starts the FastAPI server.	■ GET /health ■ health_check	115
prompts.py backend/app/prompts.py	LLM system-prompt templates used by the orchestrator and code-gen pipeline.	—	352
__init__.py backend/app/routes/__init__.py	Python module providing supporting logic for the service.	—	1
agents.py backend/app/routes/agents.py	FastAPI router exposing CRUD endpoints for Mistral agent resources.	■ GET /agents ■ list_agents ■ GET /agents/{agent_id} ■ get_agent ■ POST /agents ■ create_agent ■ PATCH /agents/{agent_id} ■ update_agent ■ DELETE /agents/{agent_id} ■ delete_agent	57
chat.py backend/app/routes/chat.py	FastAPI router for streaming general-purpose chat completions.	■ POST /chat/completions ■ chat_completions ■ POST /chat/stream ■ chat_stream	63
conversations.py backend/app/routes/conversations.py	FastAPI router for listing and deleting conversation histories.	■ GET /conversations ■ list_conversations ■ GET /conversations/{conversation_id} ■ get_conversation ■ GET /conversations/{conversation_id}/history ■ get_conversation_history ■ DELETE /conversations/{conversation_id} ■ delete_conversation	35
orchestrator.py backend/app/routes/orchestrator.py	FastAPI router for the multi-agent orchestrator chat interface.	■ POST /orchestrate ■ orchestrate_endpoint ■ POST /orchestrate/stream ■ orchestrate_stream_endpoint	50

File	Description	APIs / Endpoints	LOC
tools.py backend/app/routes/tools.py	FastAPI router for tool CRUD, synthesis triggering, and execution.	■ GET /tools/pending ■ <i>get_pending_tools</i> ■ POST /tools/{tool_id}/approve ■ <i>approve_tool</i> ■ POST /tools/{tool_id}/reject ■ <i>reject_tool</i> ■ DELETE /tools/{tool_id} ■ <i>delete_tool</i> ■ PUT /tools/{tool_id} ■ <i>update_tool</i> ■ GET /tools ■ <i>list_all_tools</i> ■ GET /tools/by-hash/{hash} ■ <i>get_tool_by_hash</i> ■ POST /tools/synthesize ■ <i>synthesize_tool</i> ■ GET /mcp/servers ■ <i>list_mcp_servers</i> ■ POST /mcp/servers ■ <i>register_mcp_server</i> ■ POST /mcp/execute/{server_name}/{tool_name} ■ <i>execute_mcp_tool</i> ■ POST /mcp/health-check ■ <i>mcp_health_check</i>	133

File	Description	APIs / Endpoints	LOC
workflows.py backend/app/routes/workflows.py	FastAPI router for workflow planning, execution, archiving, and history.	■ POST /workflows/plan ■ plan_workflow_endpoint ■ POST /workflows ■ create_workflow ■ GET /workflows ■ list_all_workflows ■ GET /workflows/{workflow_name} ■ get_workflow_detail ■ PUT /workflows/{workflow_name}/archive ■ archive_workflow_endpoint ■ PUT /workflows/{workflow_name}/unarchive ■ unarchive_workflow_endpoint ■ POST /workflows/{workflow_name}/execute ■ execute_workflow_endpoint ■ GET /workflows/executions/{execution_id} ■ get_execution_status ■ GET /workflows/executions/{execution_id}/stream ■ stream_execution_status ■ POST /workflows/executions/{execution_id}/signal ■ send_signal_to_execution ■ GET /workflows/{workflow_name}/executions ■ get_workflow_executions ■ POST /workflows/{workflow_name}/register ■ register_workflow_on_mistral ■ POST /workflows/{workflow_name}/export ■ export_to_mistral	836
__init__.py backend/app/services/__init__.py	Python module providing supporting logic for the service.	—	1
_worker_inner.py backend/app/services/_worker_inner.py	Inner subprocess logic for the Mistral Temporal workflow worker.	—	254
agent_service.py backend/app/services/agent_service.py	Business logic for creating, listing, and deleting Mistral agents via SDK.	—	144
chat_service.py backend/app/services/chat_service.py	Handles streaming chat completions using the Mistral SDK.	—	193

File	Description	APIs / Endpoints	LOC
conversation_service.py backend/app/services/conversation_service.py	CRUD operations on conversation records stored in the database.	—	95
conversational_workflow_service.py backend/app/services/conversational_workflow_service.py	Orchestrates multi-turn conversational workflow execution sessions.	—	190
mistral_worker.py backend/app/services/mistral_worker.py	Manages the background Temporal worker process lifecycle.	—	193
mistral_workflows_compiler.py backend/app/services/mistral_workflows_compiler.py	Compiles JSON workflow plans into executable Mistral Workflow Python modules.	—	177
orchestrator_service.py backend/app/services/orchestrator_service.py	Multi-agent orchestration logic: selects, invokes, and chains agent calls.	—	630
tool_registry.py backend/app/services/tool_registry.py	Maintains an in-memory registry of available tools and their metadata.	—	183
tool_resolver.py backend/app/services/tool_resolver.py	Resolves tool call requests to the appropriate executor (native, dynamic, MCP).	—	227
workflow_planner.py backend/app/services/workflow_planner.py	LLM-driven planner that converts user goals into structured workflow plans.	—	320
__init__.py backend/app/services/workflow_engine/__init__.py	Python module providing supporting logic for the service.	—	1
engine.py backend/app/services/workflow_engine/engine.py	Core workflow execution engine: step scheduling, retries, and state tracking.	—	217

File	Description	APIs / Endpoints	LOC
models.py backend/app/services/workflow_engine/models.py	SQLAlchemy ORM model definitions for database tables.	—	109
step_runners.py backend/app/services/workflow_engine/step_runners.py	Concrete runners for each workflow step type (agent, tool, conditional, etc.).	—	370

■ Tool Service (FastAPI)

File	Description	APIs / Endpoints	LOC
Dockerfile tool-service/Dockerfile	Docker image build instructions for containerising the service.	—	25
docker-compose.yml tool-service/docker-compose.yml	Multi-container Docker Compose configuration (app + DB services).	—	44
requirements.txt tool-service/requirements.txt	Python package dependencies for the service.	—	7
sandbox_requirements.txt tool-service/sandbox_requirements.txt	Additional Python packages installed inside the sandboxed tool executor.	—	7
__init__.py tool-service/app/__init__.py	Python module providing supporting logic for the service.	—	1
config.py tool-service/app/config.py	Loads and exposes environment-level configuration (API keys, DB URL, etc.).	—	31
database.py tool-service/app/database.py	SQLAlchemy engine / session factory and Base declarative model.	—	35
main.py tool-service/app/main.py	Application entry point; registers routers and starts the FastAPI server.	■ GET /health ■ health_check	87

File	Description	APIs / Endpoints	LOC
models.py tool-service/app/models.py	SQLAlchemy ORM model definitions for database tables.	—	23
prompts.py tool-service/app/prompts.py	LLM system-prompt templates used by the orchestrator and code-gen pipeline.	—	117
schemas.py tool-service/app/schemas.py	Pydantic request/response schema definitions.	—	74
seed_native_tools.py tool-service/app/seed_native_tools.py	Seeds the database with pre-built native tool definitions on startup.	—	454
__init__.py tool-service/app/routes/__init__.py	Python module providing supporting logic for the service.	—	1
execution.py tool-service/app/routes/execution.py	Tool-service route for executing a specific tool by ID.	■ POST /execute/{tool_name} ■ <i>execute</i>	26
management.py tool-service/app/routes/management.py	Tool-service routes for registering, listing, updating, and deleting tools.	■ GET /tools ■ <i>list_tools</i> ■ GET /tools/by-hash/{hash} ■ <i>get_tool_by_hash</i> ■ GET /tools/pending ■ <i>list_pending_tools</i> ■ POST /tools/{tool_id}/approve ■ <i>approve</i> ■ POST /tools/{tool_id}/reject ■ <i>reject</i> ■ DELETE /tools/{tool_id} ■ <i>delete</i> ■ PUT /tools/{tool_id} ■ <i>update</i>	89

File	Description	APIs / Endpoints	LOC
mcp.py tool-service/app/routes/mcp.py	Tool-service route for proxying MCP server tool calls.	■ GET /servers ■ list_mcp_servers ■ POST /servers ■ register_mcp_server ■ POST /execute/{server_name}/{tool_name} ■ execute_mcp_tool ■ POST /health-check ■ health_check_all	59
synthesis.py tool-service/app/routes/synthesis.py	Tool-service route for triggering AI-driven tool code synthesis.	■ POST /synthesize ■ synthesize	31
__init__.py tool-service/app/services/__init__.py	Python module providing supporting logic for the service.	—	1
execution_service.py tool-service/app/services/execution_service.py	Safely executes dynamic tool code inside a restricted sandbox environment.	—	186
llm_fallback.py tool-service/app/services/llm_fallback.py	Provides LLM-generated synthetic fallback results when tool execution fails.	—	123
mcp_manager.py tool-service/app/services/mcp_manager.py	Manages connections to Model Context Protocol servers and proxies tool calls.	—	213
sandbox.py tool-service/app/services/sandbox.py	Configures and enforces the Python sandbox for untrusted tool execution.	—	80
synthesis_service.py tool-service/app/services/synthesis_service.py	Generates Python tool source code from a natural-language description via LLM.	—	436

■■ Frontend (React + TypeScript)

File	Description	APIs / Endpoints	LOC
DESIGN.md frontend/DESIGN.md	Frontend design-system specification (colours, typography, component patterns).	—	121
README.md frontend/README.md	Project-level README with setup, architecture overview, and usage instructions.	—	73
eslint.config.js frontend/eslint.config.js	ESLint linting rules for the TypeScript/React codebase.	—	22
index.html frontend/index.html	Root HTML shell loaded by Vite; mounts the React SPA.	—	17
package.json frontend/package.json	NPM package manifest: dependencies, scripts, and project metadata.	—	48
tsconfig.app.json frontend/tsconfig.app.json	TypeScript configuration for the application source.	—	25
tsconfig.json frontend/tsconfig.json	Root TypeScript compiler configuration.	—	7
tsconfig.node.json frontend/tsconfig.node.json	TypeScript configuration for Node build scripts.	—	24
vite.config.ts frontend/vite.config.ts	Vite bundler configuration for the React/TypeScript frontend.	—	14
App.tsx frontend/src/App.tsx	Root React component; defines client-side routing and global layout.	—	66
index.css frontend/src/index.css	Global CSS reset and base styles for the frontend.	—	176
main.tsx frontend/src/main.tsx	React DOM entry point; renders into the HTML shell.	—	14
agents.ts frontend/src/api/agents.ts	Frontend API calls for agent CRUD operations.	■ GET /api/agents ■ agentsApi	28

File	Description	APIs / Endpoints	LOC
chat.ts frontend/src/api/chat.ts	Frontend API calls for general chat streaming.	■ POST /api/chat/completions ■ chatApi	8
client.ts frontend/src/api/client.ts	Configured Axios HTTP client with base URL and auth interceptors.	■ ? — ■ api	28
conversations.ts frontend/src/api/conversations.ts	Frontend API calls for conversation management.	■ GET /api/conversations ■ conversationsApi	8
health.ts frontend/src/api/health.ts	Frontend API call to check backend liveness.	■ GET /health ■ healthApi	5
mcp.ts frontend/src/api/mcp.ts	Frontend API calls for MCP server tool queries.	■ GET /api/mcp/servers ■ mcpApi	9
orchestrator.ts frontend/src/api/orchestrator.ts	Frontend API calls for the orchestrator chat interface.	■ POST /api/orchestrate ■ orchestratorApi	27
sse.ts frontend/src/api/sse.ts	Server-Sent Events helper for real-time streaming responses from the backend.	■ ? — ■ createSSEStream	67
tools.ts frontend/src/api/tools.ts	TypeScript/React module providing UI or API-client functionality.	■ GET /api/tools ■ toolsApi	12
workflowPlanner.ts frontend/src/api/workflowPlanner.ts	Frontend API calls for workflow planning requests.	■ ? /api/workflows/plan ■ workflowPlannerApi	33
workflows.ts frontend/src/api/workflows.ts	Frontend API calls for workflow lifecycle management.	■ GET /api/workflows ■ workflowsApi	30
AppShell.tsx frontend/src/components/layout/AppShell.tsx	TypeScript/React module providing UI or API-client functionality.	—	25
Sidebar.tsx frontend/src/components/layout/Sidebar.tsx	TypeScript/React module providing UI or API-client functionality.	—	348

File	Description	APIs / Endpoints	LOC
AgentDetail.tsx frontend/src/features/agents/AgentDetail.tsx	Page displaying a single agent's configuration and conversation interface.	—	323
AgentStudio.tsx frontend/src/features/agents/AgentStudio.tsx	Agent Studio page: lists, creates, and manages Mistral agents.	—	218
GeneralChat.tsx frontend/src/features/chat/GeneralChat.tsx	General-purpose chat page with streaming message support.	—	190
OrchestratorChat.tsx frontend/src/features/chat/OrchestratorChat.tsx	Orchestrator chat page that shows multi-agent reasoning steps.	—	281
ConversationManager.tsx frontend/src/features/conversations/ConversationManager.tsx	TypeScript/React module providing UI or API-client functionality.	—	102
HealthDashboard.tsx frontend/src/features/health/HealthDashboard.tsx	TypeScript/React module providing UI or API-client functionality.	—	139
McpRegistry.tsx frontend/src/features/mcp/McpRegistry.tsx	TypeScript/React module providing UI or API-client functionality.	—	180
ToolLifecycle.tsx frontend/src/features/tools/ToolLifecycle.tsx	Tool Lifecycle page: tool creation, synthesis, editing, and deletion.	—	439
ArchivedWorkflows.tsx frontend/src/features/workflows/ArchivedWorkflows.tsx	Displays the list of archived (completed) workflow executions.	—	160
WorkflowDashboard.tsx frontend/src/features/workflows/WorkflowDashboard.tsx	Workflow Dashboard: summary cards and navigation hub for workflows.	—	292

File	Description	APIs / Endpoints	LOC
WorkflowExecutionModal.tsx frontend/src/features/workflows/WorkflowExecutionModal.tsx	Modal dialog for launching and configuring a workflow execution.	—	671
WorkflowExecutionPage.tsx frontend/src/features/workflows/WorkflowExecutionPage.tsx	Full-page real-time workflow execution view with live step tracking.	—	722
WorkflowHistoryPanel.tsx frontend/src/features/workflows/WorkflowHistoryPanel.tsx	Side panel showing the execution history for a given workflow.	—	95
WorkflowPlanner.tsx frontend/src/features/workflows/WorkflowPlanner.tsx	Interactive UI for designing and submitting new workflow plans.	—	461
WorkflowVisualizer.tsx frontend/src/features/workflows/WorkflowVisualizer.tsx	SVG-based visual graph of workflow steps and their dependencies.	—	608
queryClient.ts frontend/src/lib/queryClient.ts	TypeScript/React module providing UI or API-client functionality.	—	27
utils.ts frontend/src/lib/utils.ts	TypeScript/React module providing UI or API-client functionality.	—	5
chatStore.ts frontend/src/store/chatStore.ts	TypeScript/React module providing UI or API-client functionality.	—	56
sessionStore.ts frontend/src/store/sessionStore.ts	TypeScript/React module providing UI or API-client functionality.	—	176
tokens.css frontend/src/styles/tokens.css	CSS stylesheet.	—	34

■ Mistral Workflows

File	Description	APIs / Endpoints	LOC
workflow_personalized_fitness_plan_workflow.py mistral_workflows/workflow_personalized_fitness_plan_workflow.py	Compiled Mistral workflow that generates a personalised fitness plan using agent steps.	—	217

■ Summary

Total Files	Total Lines of Code	Total API Endpoints
113	18,202	64
files analysed	across all source files	routes & client calls found